

Physical Interaction for Aerial and Space Robots (AE4324)

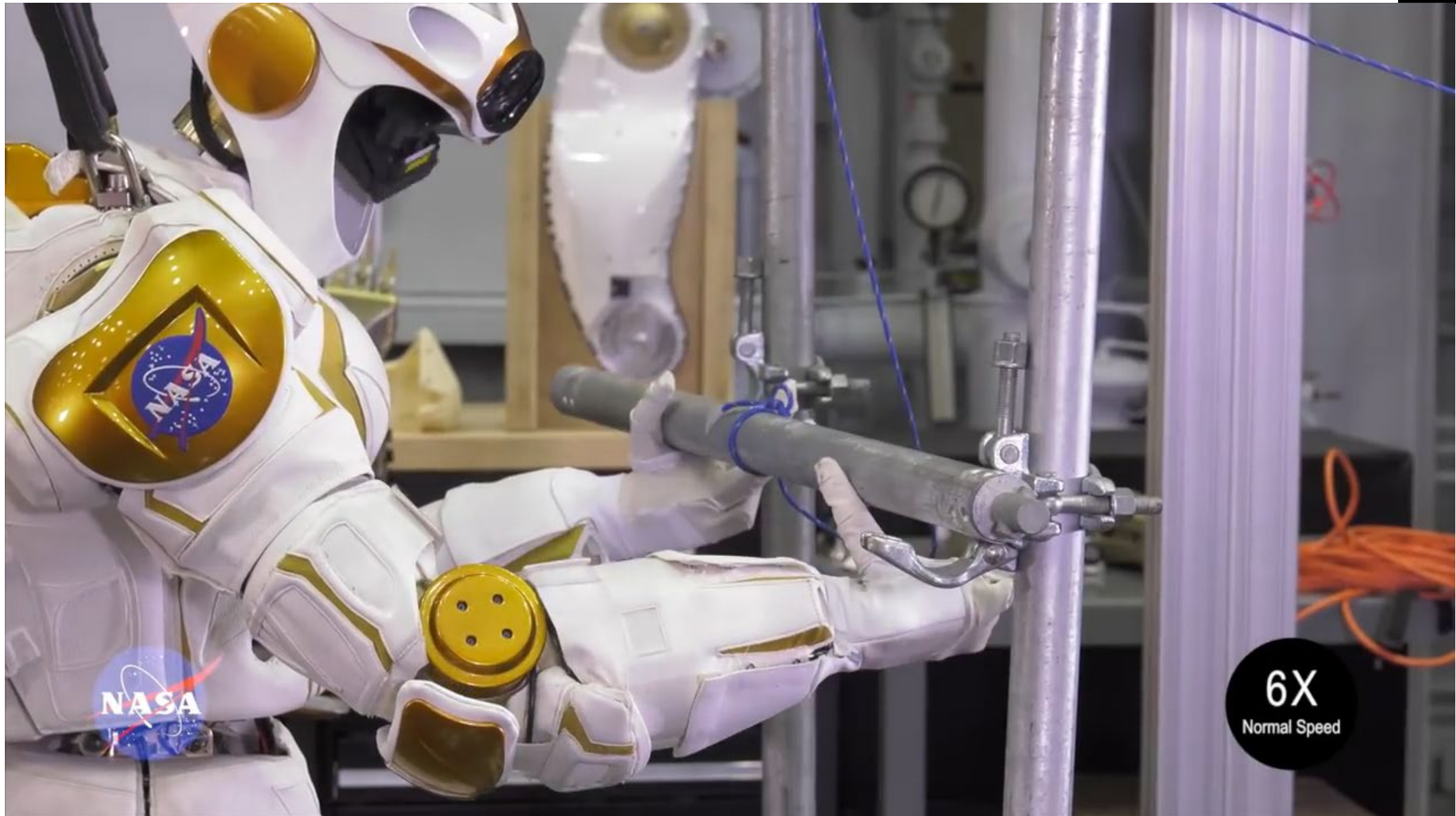
Lecture 8: Physical Interaction II



Today's Learning Objectives

1. Evaluate compliant control methods
2. Apply Impedance /Admittance / Hybrid Position-Force Control
3. Evaluate Passivity-based Control via Energy Tanks

Compliant Control



- **Stiffness Control**: can adjust the elasticity or rigidity of a system to manage how it reacts to external forces
- **Impedance Control**: regulates the relationship between position errors and interaction forces, effectively acting as a mass-spring-damper system. It is generally used with torque-controlled robots.
- **Admittance Control**: is a type of motion control that calculates required position adjustments based on measured external forces. It is often applied to position-controlled robots.
- **Hybrid Force/Position Control**: splits control into orthogonal directions, where position is controlled in directions tangent to a surface, and force is controlled normal to the surface.

Stiffness Control

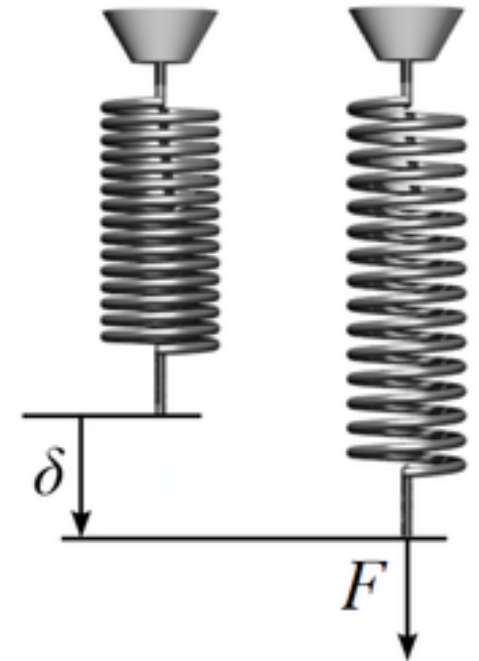
Stiffness is...

the extent to which an object resists deformation in response to an applied force.

- controls the apparent stiffness of the robot end-effector
- achieves a desired **static** behavior (disregards inertial effects)
- acts like a spring

$$\mathbf{F}_{\text{ext}} = \mathbf{K} \delta \mathbf{x}_{\text{ext}}$$

force wrench 6 x 6 desired stiffness matrix (diagonal) differential displacement



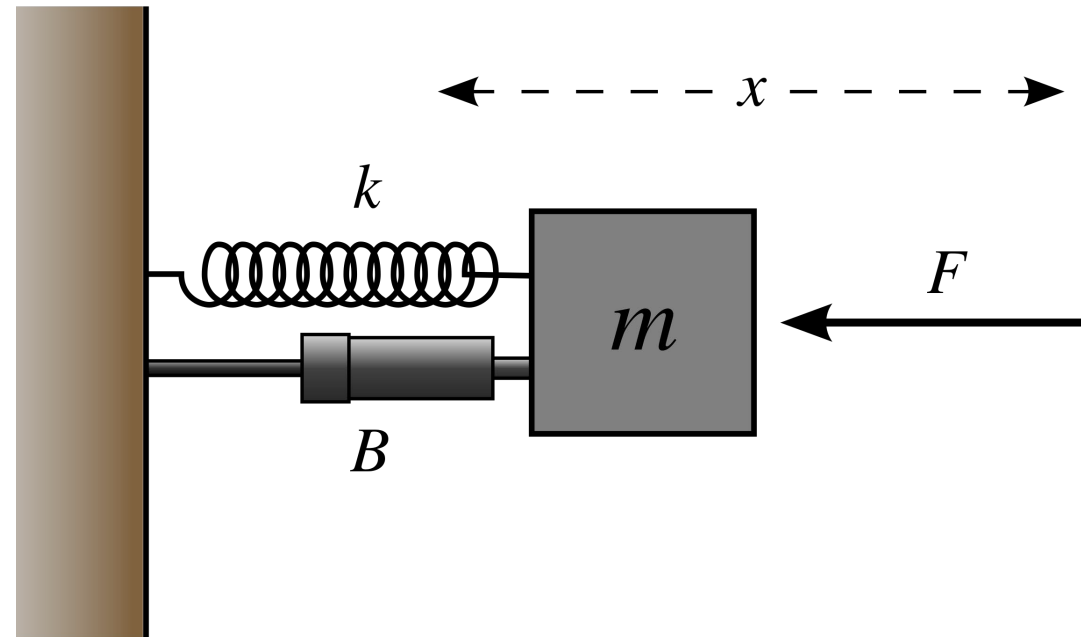
Wrench: is the spatial representation of the force/load acting on a rigid body. It combines a linear force vector and an angular torque (moment) vector into a single 6-dimensional vector. $\mathbf{F} = \begin{pmatrix} \mathbf{f} \\ \mathbf{m} \end{pmatrix}$

Impedance Control

- Impedance control is an approach to **dynamic control** relating force and position

What is mechanical impedance?

... it is a measure of how much a system **resists motion when subjected to a force**



Impedance Control

Describing the robot as a **mass-spring-damper system**, we can express the relationship between force and velocity (i.e. the impedance)

$$F_{ext} = M_m \ddot{x} + D_m \dot{x} + K_m x$$

external forces
from the
environment

apparent inertia
(> 0)

desired
damping (≥ 0)

desired
stiffness (> 0)

In Joint space

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau_m + \tau_{ext}$$

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = u + J^T(q)F_{ext}$$

In Cartesian space

$$M_x(q)\ddot{x} + C_x(q, \dot{x})\dot{x} + G_x(q) = J^{-T}(q)u + F_{ext}$$

with

$$M_x(q) = J^{-T}(q) M(q) J^{-T}(q)$$

$$C_x(q, \dot{x}) = J^{-T}(q) C(q, \dot{q}) J^{-1}(q) - M_x(q)\dot{J}(q)J^{-1}(q)$$

$$G_x(q) = J^{-T}(q)G(q)$$

the reason why we use the Cartesian dynamic model is because now the generalised forces on the right side of the equation appear directly, without any multiplication by the Jacobian

Impedance Control

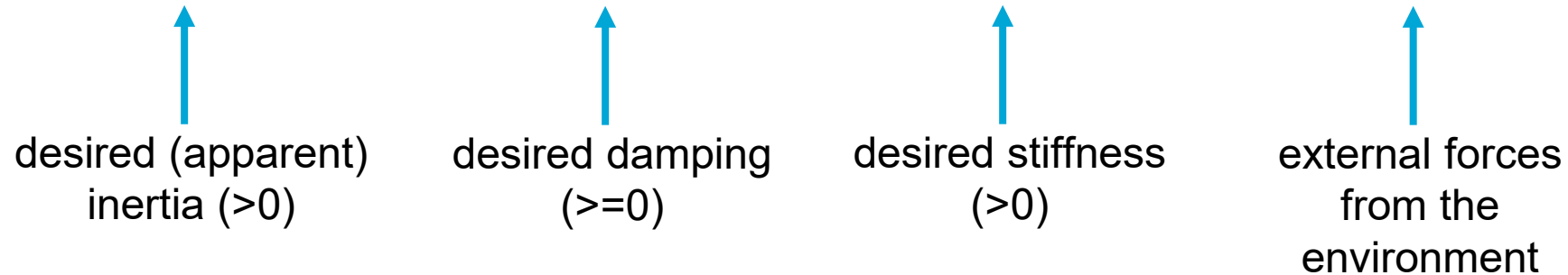
Designing the control law is based on 2 steps:

1. feedback linearisation in Cartesian space (with **force measure**)

$$u = J^T(q)[M_x(q)\ddot{x} + C_x(q, \dot{q})\dot{x} + G_x(q) - F_{ext}]$$

2. imposition of a dynamic **impedance model**

$$M_m(\ddot{x} - \ddot{x}_{des}) + D_m(\dot{x} - \dot{x}_{des}) + K_m(x - x_{des}) = F_{ext}$$



- most of the times, the matrices that multiply the EE coordinates are chosen as diagonal
- $x_{des}(t)$ is the desired motion $\rightarrow$ typically 'slightly over/inside' the object, inducing contact forces

$$u = M(q)J^{-1}(q)\{\ddot{x}_{des} - \dot{J}(q)\dot{q} + M_m^{-1} [D_m(\dot{x}_{des} - \dot{x}) + K_m(x_{des} - x)]\} \\ + C(q, \dot{q})\dot{q} + G(q) + J^T(q) \boxed{[M_x(q)M_m^{-1} - I] F_{ext}}$$

- we chose an impedance model and a reference position or trajectory slightly inside the environment
- we get rid of all Cartesian terms, we re-write them in joint space terms $M(q)$, $C(q, \dot{q})$, $G(q)$
- the first line of the eq. we have the inertia of the robot, the inverse of the Jacobian (assuming a square Jacobian), the desired acceleration coming from the impedance model, a term related to the derivative of the Jacobian and a PD term coming from the impedance model
- the term pre-multiplied by the Jacobian transpose is the measured contact forces $\rightarrow$ in this general case we assume to measure contact forces F_{ext} - e.g. by means of a force sensor at the end-effector
- lastly, by substituting the cartesian inertia M_x from the definition in joint space we get

$$J^T(q)[M_x(q)M_m^{-1} - I]F_{ext} = [M(q)J^{-1}(q)M_m^{-1} - J^T(q)]F_{ext}$$

which eliminates from the control law the last Cartesian quantity, i.e. Cartesian inertia matrix

Impedance Control in Joint Space

Choose the **apparent inertia** equal to the **natural cartesian inertia** of the robot

$$M_m = M_x(q) = J^{-T}(q)M(q)J^{-1}(q)$$

$$u = M(q)J^{-1}(q)\{\ddot{x}_{des} - \dot{J}(q)\dot{q} + M_m^{-1} [D_m(\dot{x}_{des} - \dot{x}) + K_m(x_{des} - x)]\} \\ + C(q, \dot{q})\dot{q} + G(q) + \cancel{J^T(q)[M_x(q)M_m^{-1} - I]F_{ext}}$$



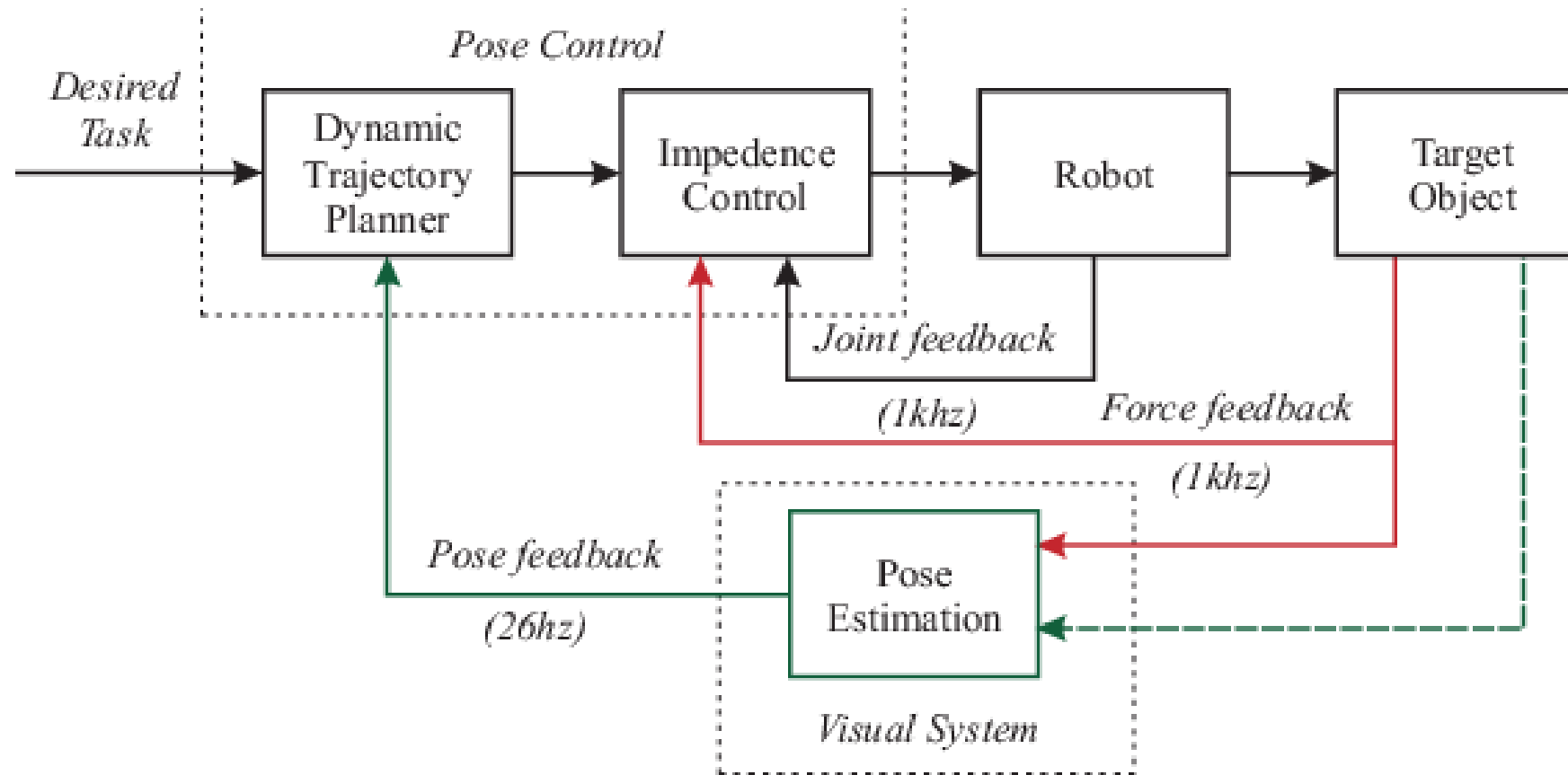
$$u = M(q)J^{-1}(q)\{\ddot{x}_{des} - \dot{J}(q)\dot{q}\} + J^T(q)[D_m(\dot{x}_{des} - \dot{x}) + K_m(x_{des} - x)] \\ + C(q, \dot{q})\dot{q} + G(q)$$

WITHOUT contact force feedback (a F/T sensor is no longer needed)

this is a pure motion control law applied also during interaction,

but designed so as to keep limited contact forces at the end-effector level (with a '*feeling*' for the environment stiffness)

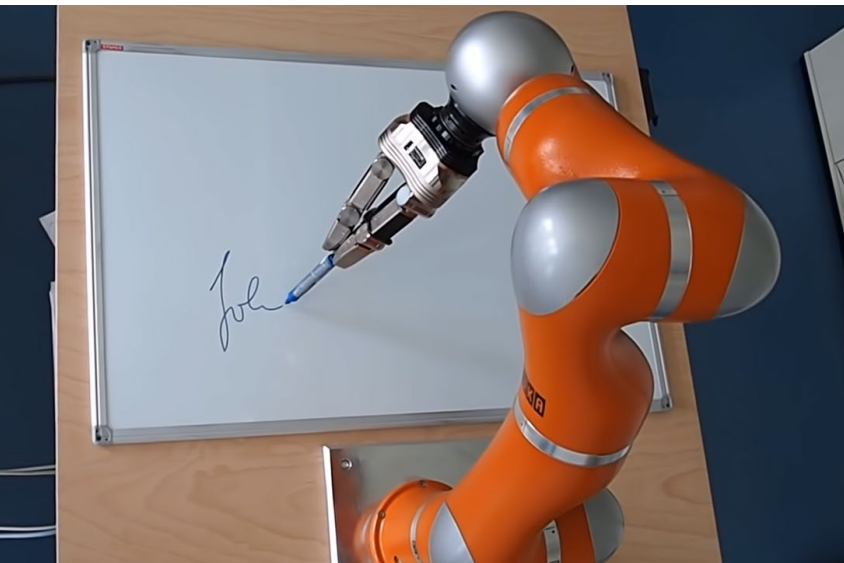
Impedance Control – Block Diagram



Impedance Control - Applications

$$M_m(\ddot{x} - \ddot{x}_{des}) + D_m(\dot{x} - \dot{x}_{des}) + K_m(x - x_{des}) = F_{ext}$$

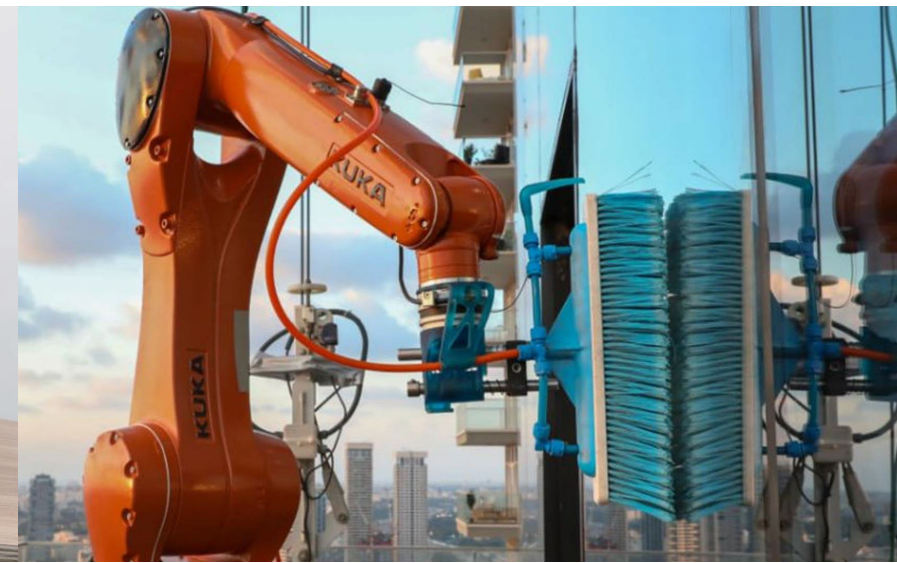
In a contact task, the desired reference is **slightly inside** the environment (ensuring continuous contact, but not too much to damage the environment)



writing



polishing



window cleaning

Impedance Control - Applications

Contact tasks where the robot goes back to a “rest position” in a human-robot interaction task

$$M_m(\ddot{x} - \ddot{x}_{des}) + D_m(\dot{x} - \dot{x}_{des}) + K_m(x - x_{des}) = F_{ext}$$



Impedance Control

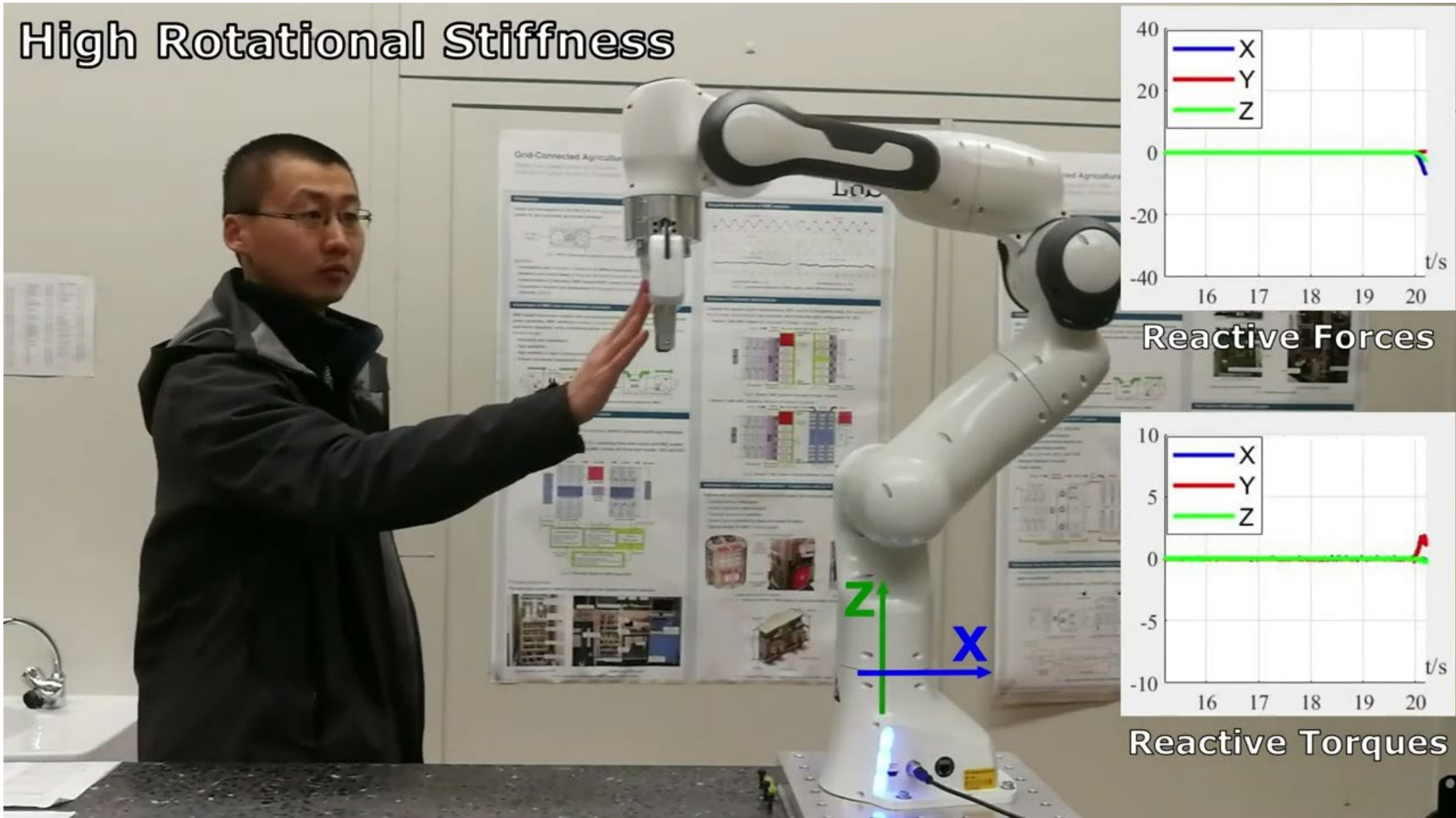


Impedance Control Summary

- imposes a desired **dynamic behaviour** to the interaction between robot end-effector and environment
- the **desired performance** is specified through a generalized dynamic impedance, i.e. **a complete set of mass-spring-damper equations** (typically chosen as linear and decoupled)
- a model describing how reaction forces are generated is **not** explicitly required
- suited for tasks in which **contact forces should be kept small**; accurate regulation is not mandatory
- No control loop on **force measurements**
- the choice of the **stiffness** in the impedance model along a Cartesian **direction** results in a **trade-off** between contact forces and position accuracy in that direction
 - the contact force can be made small at the expense of a large position error, as long as the active stiffness is set low with respect to the stiffness of the environment, and vice versa.
 - both contact force and the position error also depend on the external disturbance
 - the lower the apparent stiffness, the higher the influence of the external load
 - a low stiffness may result in a large position error also in the absence of interaction (i. e., when $k = 0$).

Impedance Control with a 7-DOF Robotic Arm

High Rotational Stiffness

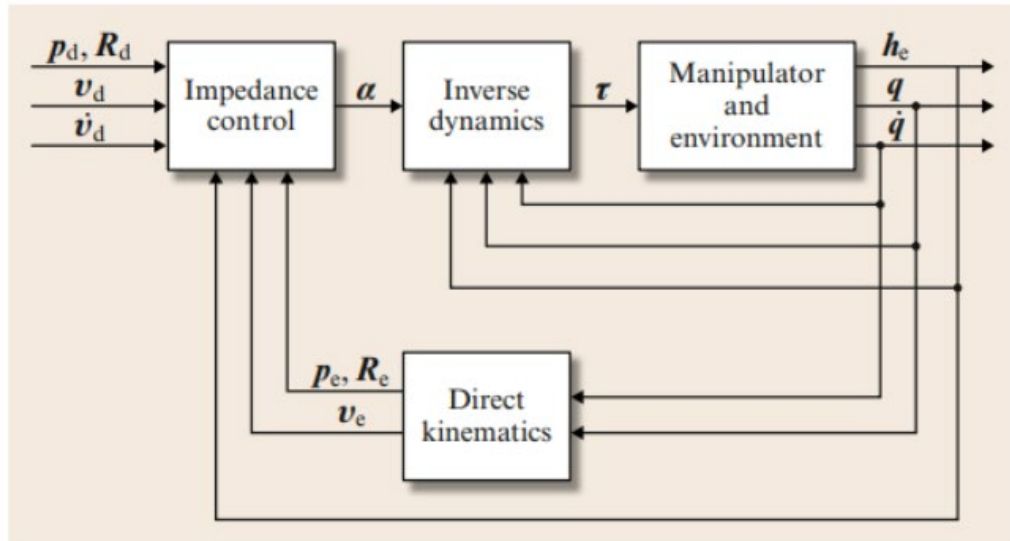


Admittance Control vs. Impedance Control

Admittance: Measures force, outputs position/velocity

Impedance: Measures position/velocity, outputs force

Impedance Control

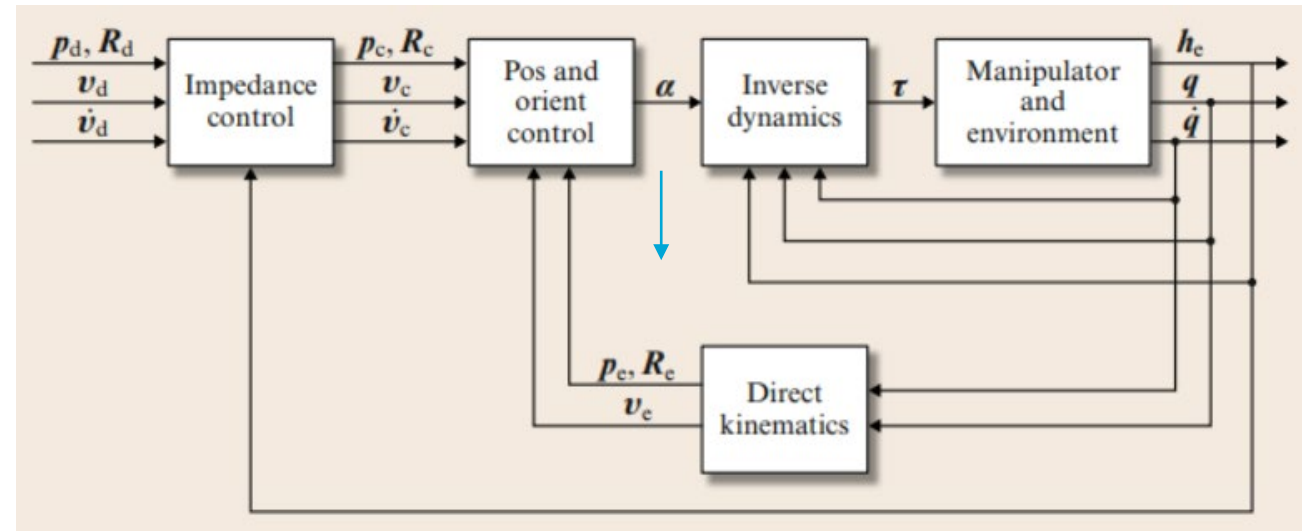


Estimated
force



Motion
(position & velocity)

Admittance Control



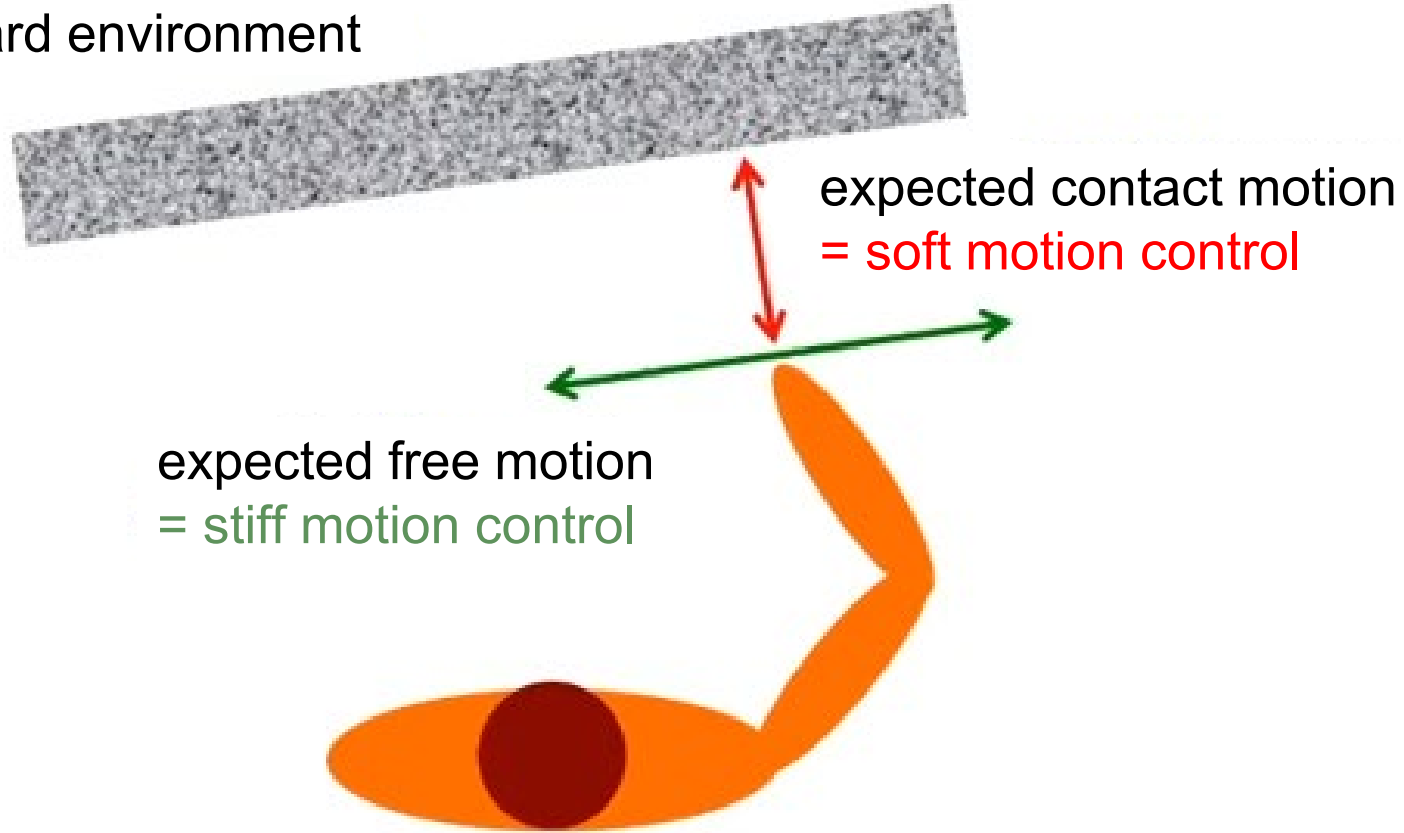
EE displacement /
velocity from
desired motion
trajectory



Contact
Wrench

Hybrid Force/Position Control

hard environment



in the selected Cartesian direction i :

the **stiffer** the environment, the **softer** is the chosen model stiffness K_{m_i}

Hybrid Force/Position Control

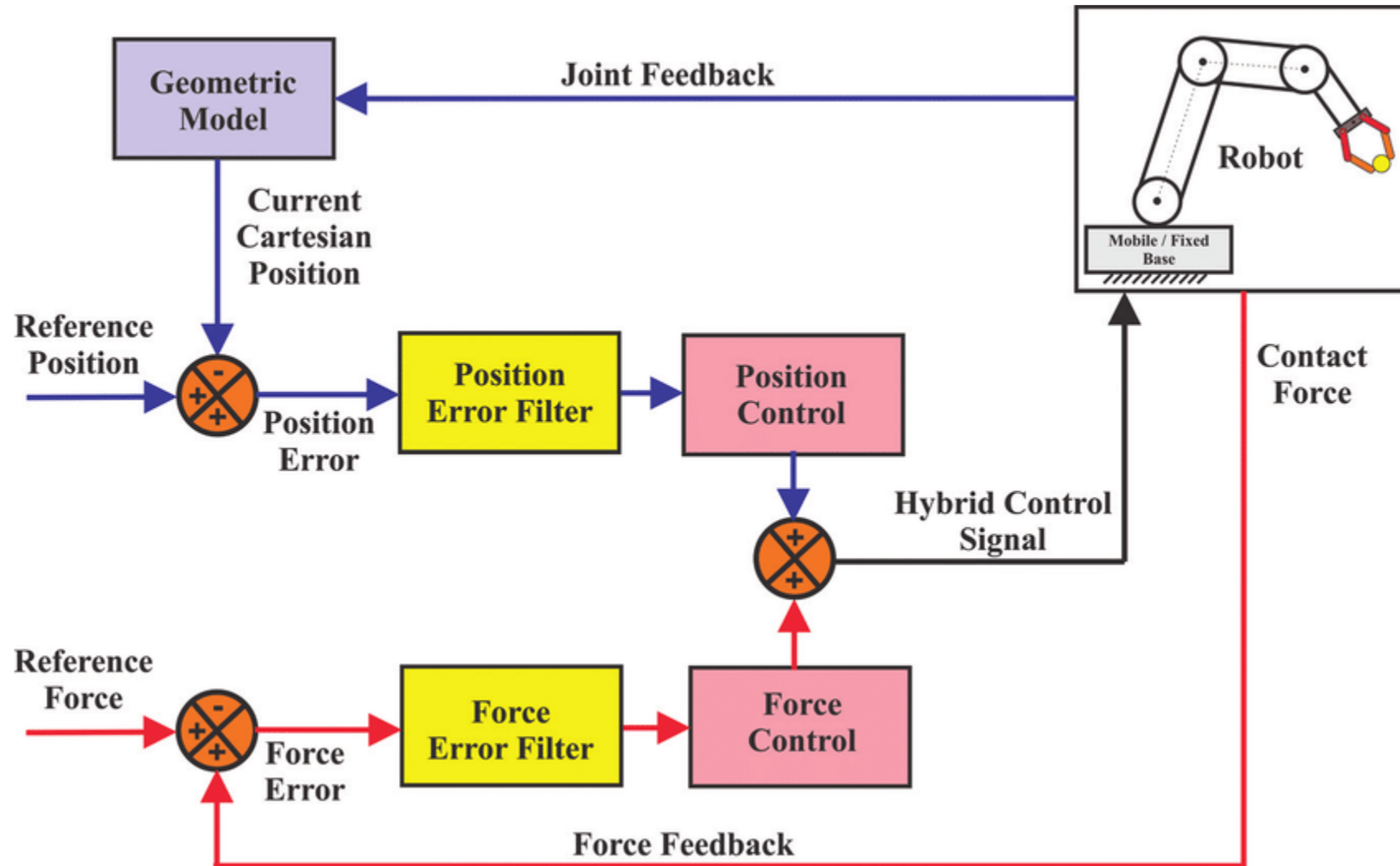
... manages complex interaction tasks by simultaneously regulating position in free directions and force in constrained directions

... It partitions the task space into orthogonal subspaces, applying distinct controllers to manage compliant contact, for example for assembly or polishing tasks, preventing excessive force.

... the desired motion satisfies the velocity constraints all the time

... we can design the two controllers independently by having a projection matrix that projects an arbitrary end-effector wrench to the portion of that wrench that cause motion of the end-effector

Hybrid Force/Position Control



Hybrid Force/Position Control

The wrench created at the end-effector is the sum of the wrench specified by the force control law, and the wrench specified by the task-space motion controller

$$F_b = (I - P(\theta)) (\textit{force controller output}) + \dots \\ + P(\theta)(\textit{task space motion controller output})$$

$$\tau = J^T(\theta)F_b$$

commanded joint torques using the Jacobian transpose

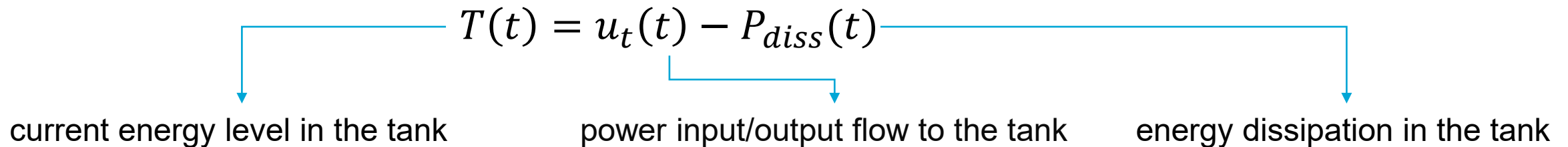
Energy-Tank-Based Control

The role of the energy tank is to act as a storage element and minimise the energy dissipated by the controlled system.

$$P_{diss}(t) = \frac{1}{2} k x^2$$

elastic potential energy dissipated

Such dissipated energy represents a **passivity threshold**, and the tank is its energy reservoir. Basically, it is a way to prescribe passive control actions (variable impedance or safety actions) by storing and releasing energy in a virtual “tank”.



To ensure passivity, the control input τ_c (e.g. in impedance control $\tau_c = D \dot{x} + K x$) is modulated based on the available energy in the tank. A common formulation uses a variable to scale the control action:

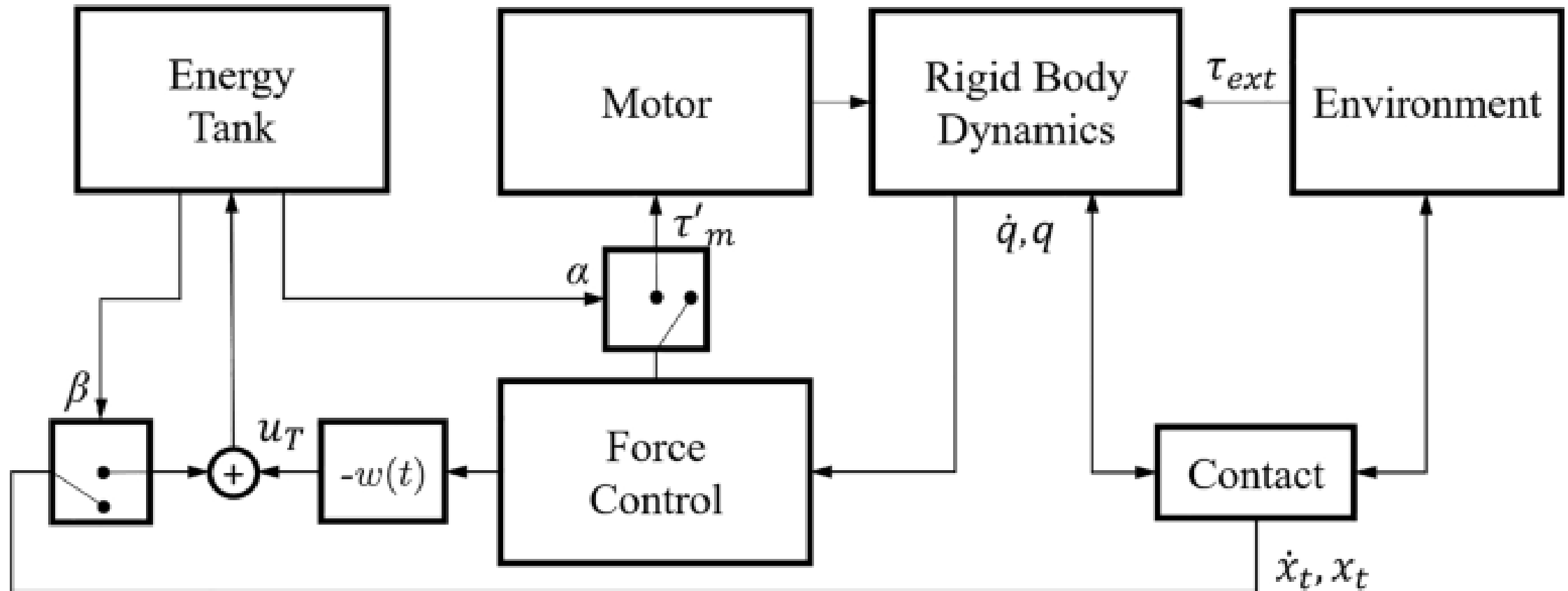
$$\tau_{mod} = \beta \tau_c \quad \text{energy modulation}$$

Where $\beta \in [0,1]$ is a modulation factor

→ $\beta = 1$ when the tank has sufficient energy $T(t) > T_{min}$

→ $\beta = 0$ when the tank is empty $T(t) \leq T_{min}$ disabling the action to prevent instability

Energy-Tank-Based Control



Summary

1. Evaluate compliant control methods
2. Apply Impedance / Admittance / Hybrid Position-Force Control
EoM, conditions, applications, examples
3. Evaluate Passivity-based Control via Energy Tanks
state and modulation + energy dissipation threshold

fin